## Supplementary Material

# Synergy of Nanoconfinement and Promotion in the Design of Efficient Supported Iron Catalysts for Direct Olefin Synthesis from Syngas

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**Conversion and selectivity calculations:**

The reaction feed and product analyses are based on gas chromatography (GC). The *A* appearing in the following calculation method indicates the peak area in the GC, *f* indicates the calibration factor, *n* means moles of products (in terms of carbon atom moles), and *x* and *y* indicate the number of carbon atoms and the number of hydrogen atoms in any one hydrocarbon and *S* shows products selectivity.

The CO conversion rate is calculated using $N_2$ (or He) as an internal standard:

$$CO_{(Conv.)\%} = \left(1 - \frac{A_{initial\ N2}/A_{reaction\ N2}}{A_{initial\ CO}/A_{reaction\ CO}}\right) * 100\% \qquad (1\text{-}1)$$

The feed gas of the internal standard gas ($N_2$ or He) is controlled by mass flow controller (MFC). The inlet molars of CO and $N_2$ ($n_{CO}$ and $n_{N2}$) can be calculated from the gas state equation (Equation 2-2). So the generation rate of $CH_4$ and $CO_2$ can be calculated by Equation 2-3 and Equation 2-4, respectively, and the selectivity can be also calculated by Equation 2-5 and Equation 2-6, respectively.

$$P*V = nRT \qquad (1\text{-}2)$$

$$n_{CH_4} = \frac{A_{CH_4}/f_{CH_4}}{A_{N2}/f_{N2}} * n_{N2} \qquad (1\text{-}3)$$

$$n_{CO_2} = \frac{A_{CO_2}/f_{CO_2}}{A_{N2}/f_{N2}} * n_{N2} \qquad (1\text{-}4)$$

$$S_{CH_4} = \frac{n_{CH_4}}{n_{CO} * CO_{(Conv.)\%} - n_{CO_2}} * 100\% \qquad (1\text{-}5)$$

$$S_{CO_2} = \frac{n_{CO_2}}{n_{CO} * CO_{(Conv.)\%}} * 100\% \qquad (1\text{-}6)$$

Due to the correlation between the two detectors, the product selectivity calculation is calculated by using $CH_4$ as the second internal standard in the product, that is, the relative amount of each product to $CH_4$ is first calculated, and the product selectivity is calculated. FID is a carbon counting device, which can be understood as a carbon atom corresponding to a unit peak area. The relative amounts and selectivity of each component of the carbon-containing compound in the products can be calculated by Equation 2-7 and 2-8, respectively.

$$n_{C_j} = \frac{A_{C_j}/f_{C_j}}{A_{CH_4}/f_{CH_4}} * n_{CH_4} \qquad (1\text{-}7)$$

$$S_{C_j} = \frac{n_{C_j}}{n_{CO} * CO_{(Conv.)\%} - n_{CO_2}} * 100\% \qquad (1\text{-}8)$$

The carbon balance is calculated by the total amount of carbon actually detected (the sum of all carbon-containing compounds detected by the FID plus the $CO_2$ detected on the TCD) divided by the total amount of carbon consumed in the reaction.

$$C_{balance} = \frac{n_{CO_2} + \sum_{j=1}^{max} n_{C_j}}{n_{CO} * CO_{(Conv.)\%}} * 100\% \qquad (1\text{-}9)$$

**Table S1.** Catalytic performance comparison in different high temperature FT catalytic systems

| Catalysts | FTY $10^{-4}$ $mol_{CO}$ $g_{Fe}^{-1}s^{-1}$ | CO Conversion / % | Reaction conditions | Reference |
|---|---|---|---|---|
| FePb/CNT-in | 23.4 | 45 | P=20 bar, T=350 $^{o}$C, $H_2$/CO=1/1, GHSV= 84 $Lg^{-1}h^{-1}$ | This work |
| FeMgK1/rGO | 17.1 | 59-62 | P=20 bar, T=340 $^{o}$C, $H_2$/CO=1/1, GHSV= 168 $Lg^{-1}h^{-1}$ | [1] |
| Fe/CNF(Na,S) | 13.0 | 6 | P=20 bar, T=340 $^{o}$C, $H_2$/CO=1/1, GHSV= 84 $Lg^{-1}h^{-1}$ | [2] |
| $Fe_3O_4$@$Fe_5C_2$ | 7.2 | 28 | P=20 bar, T=300 $^{o}$C, $H_2$/CO=1/1, GHSV= 18 $Lg^{-1}h^{-1}$ | [3] |
| FeK1/rGO | 6.5 | 58-64 | P=20 bar, T=340 $^{o}$C, $H_2$/CO=1/1, GHSV= 72 $Lg^{-1}h^{-1}$ | [4] |
| 0.6K38-Fe@C | 4.9 | 59 | P=20 bar, T=340 $^{o}$C, $H_2$/CO=1/1, GHSV= 1 $Lg^{-1}h^{-1}$ | [5] |
| Fe/CNF (Na, S) | 2.0 | 12 | P=20 bar, T=340 $^{o}$C, $H_2$/CO=1/1, GHSV=3000 $h^{-1}$ | [6] |
| Fe/$\alpha$-$Al_2O_3$(Na, S) | 0.9 | 77 | P=20 bar, T=340 $^{o}$C, $H_2$/CO=1/1GHSV= 1500 $h^{-1}$ | [7] |

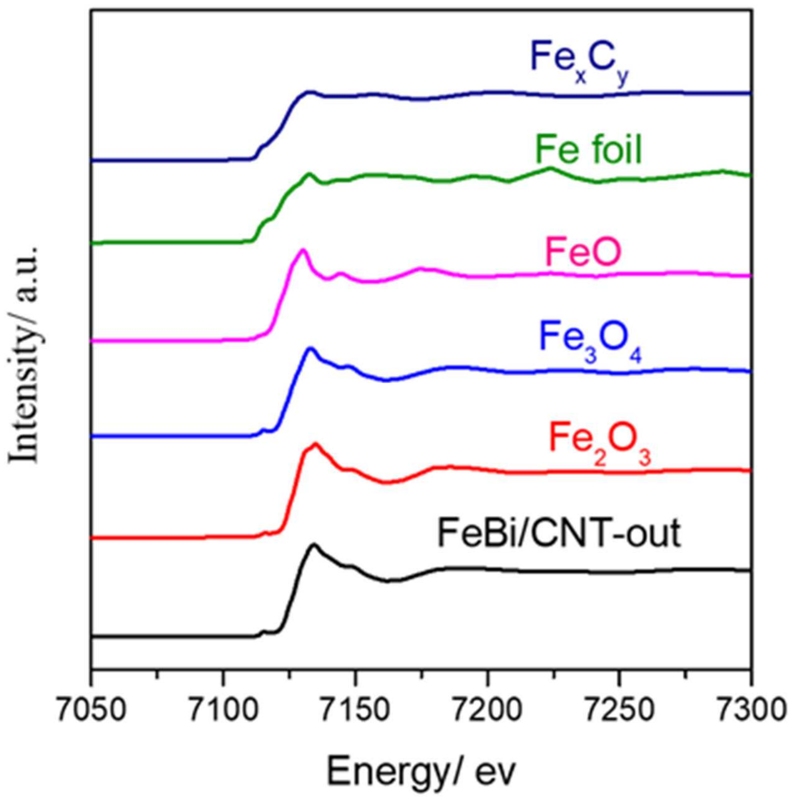


**Figure S1.** XANES spectra of iron reference compounds. The spectra are offset for clarity.

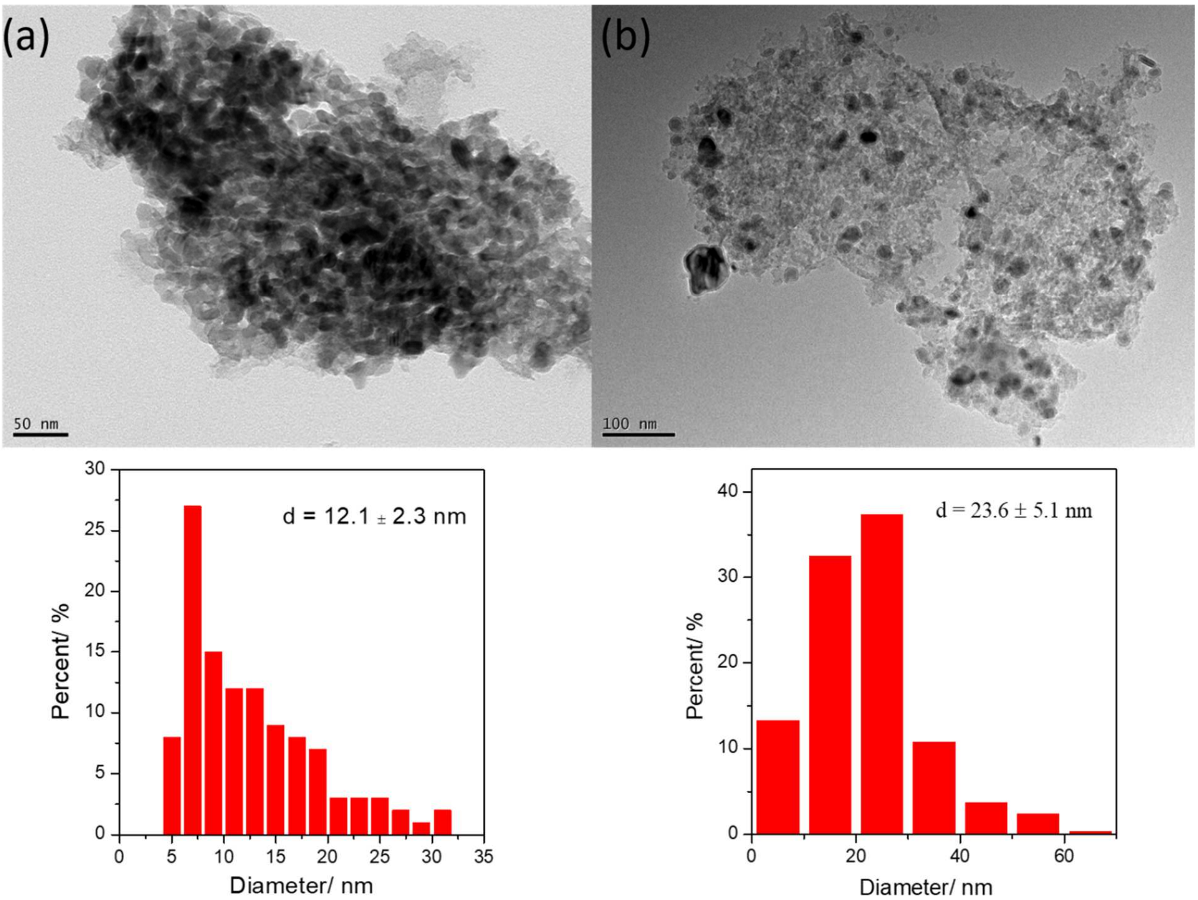


**Figure S2.** TEM micrographs and particle distribution for active carbon supported iron catalysts after: (a) fresh, (b) after reaction

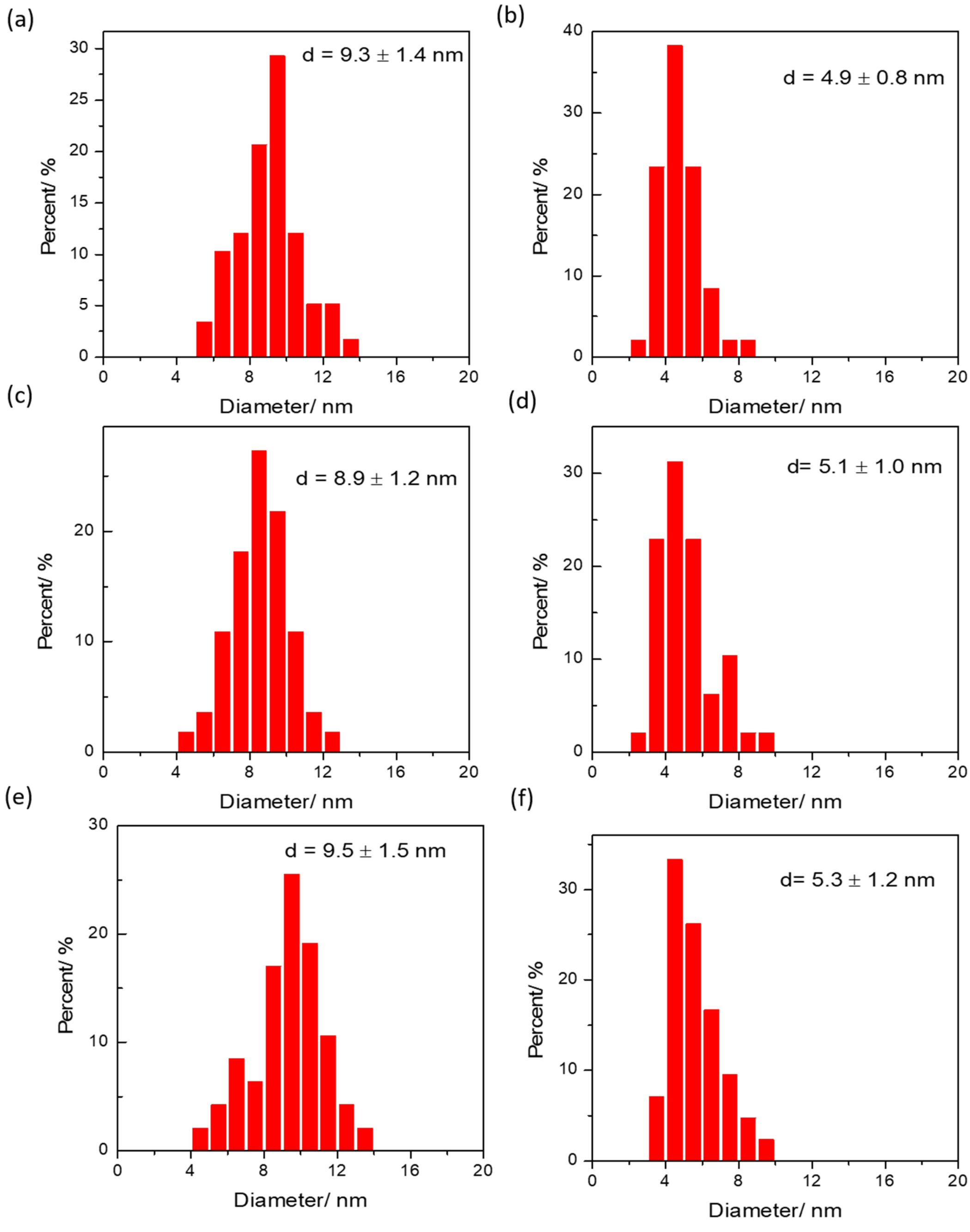


**Figure S3**. Particle size distribution for the fresh catalysts: (a) Fe/CNT-out, (b) Fe/CNT-in, (c) FeBi/CNT-out, (d) FeBi/CNT-in, (e) FePb/CNT-out, (f) FePb/CNT-in.

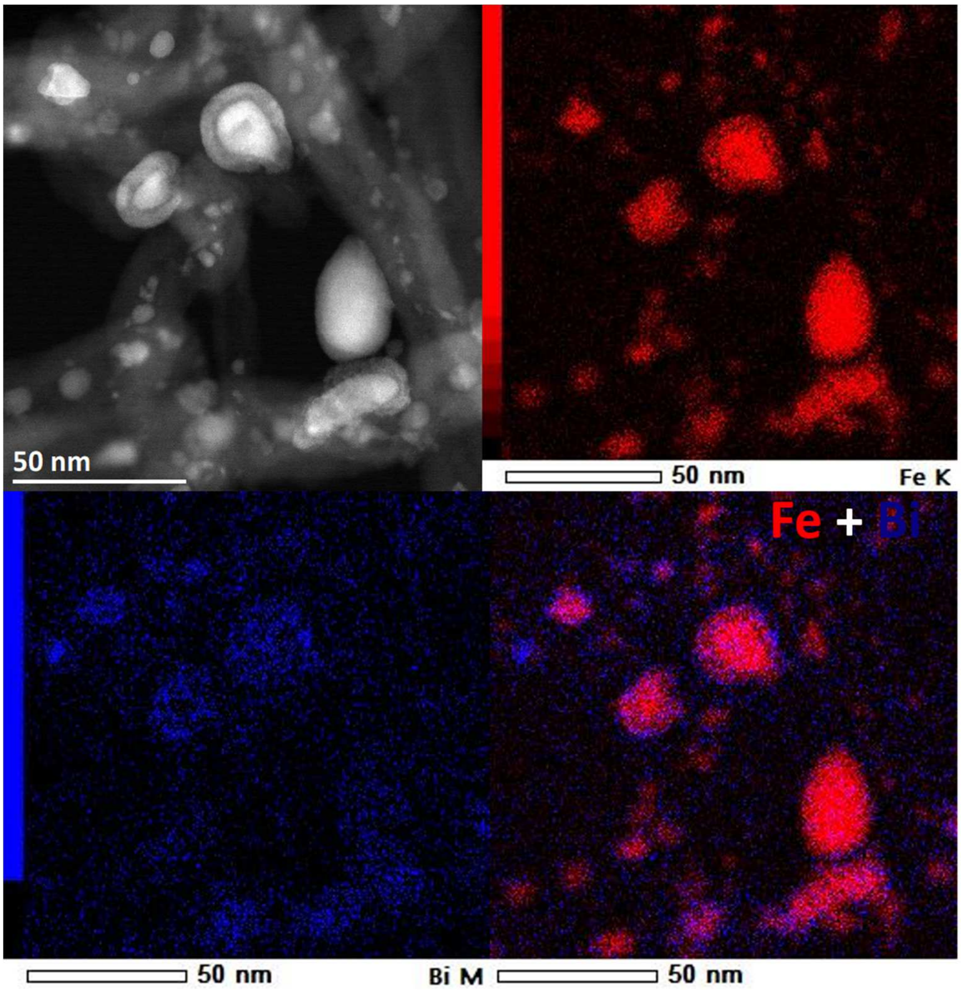


**Figure S4.** STEM-HAADF and STEM-EDS images for the activated FeBi/CNT-out catalyst (CO treatment for 10 h at 350 ºC).

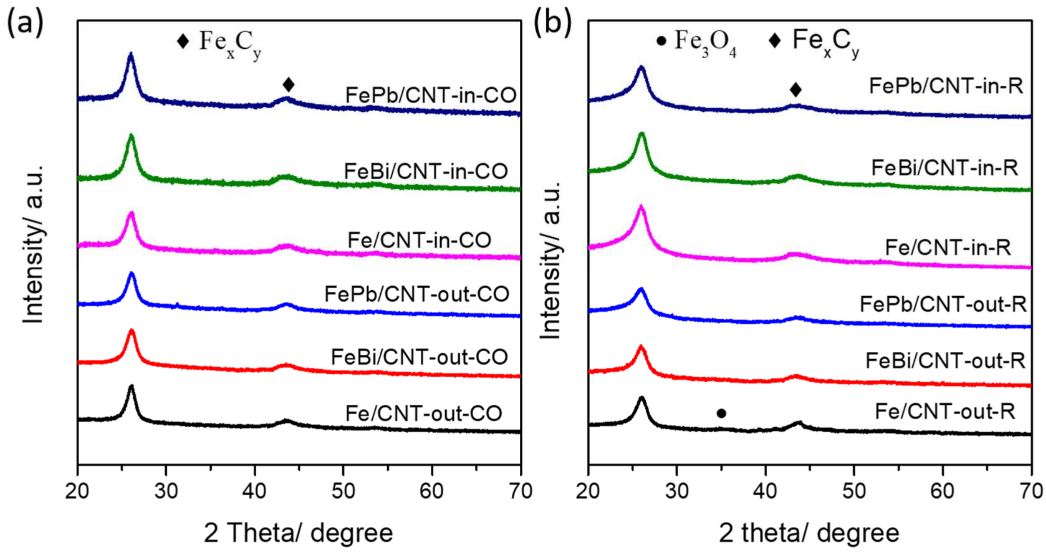


**Figure S5**. XRD profiles of the carbidized (a) catalysts after treatment at 350 ºC for 10h, (b) catalysts after reaction.

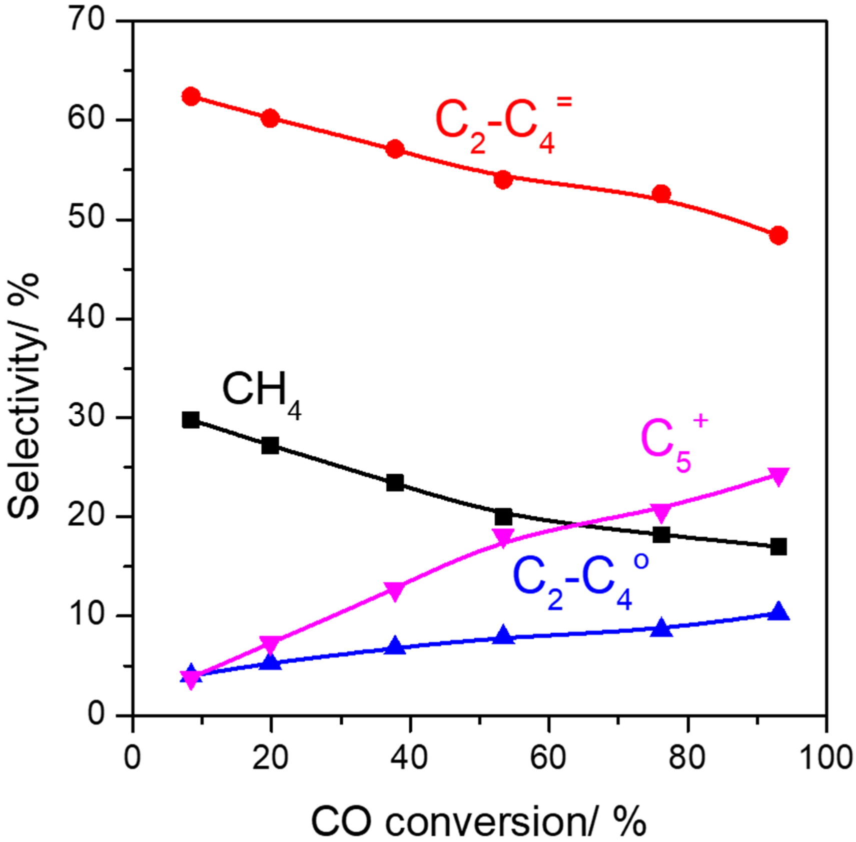


**Figure S6.** Product selectivity versus CO conversion over FePbK/CNT-in catalyst. Reaction condition: P=10 bar, $H_2$/CO=1/1, T= 350 $^o$C, GHSV= 13.8 – 68 L/g.h.

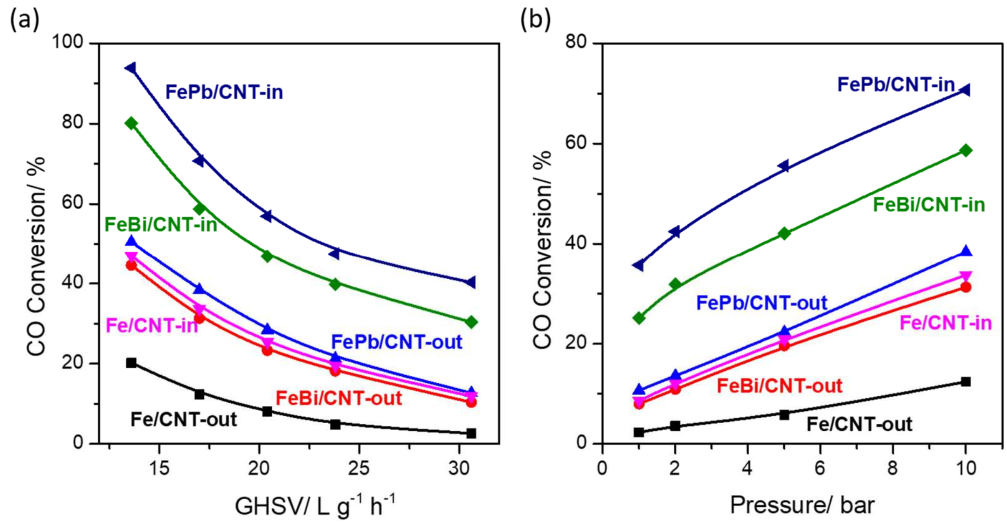


**Figure S7**. Carbon monoxide conversion over CNT confined and non-confined iron- based catalysts under different reaction condition: (a) under different GHSV, P=10 bar, (b) under different pressure, GHSV=17 L/g h. Reaction conditions: *W* = 0.2g, $H_2/CO$ = 1, *P* = 1-10 bar, T=350 $^oC$, *GHSV*= 13.6-30.6 L/g.h

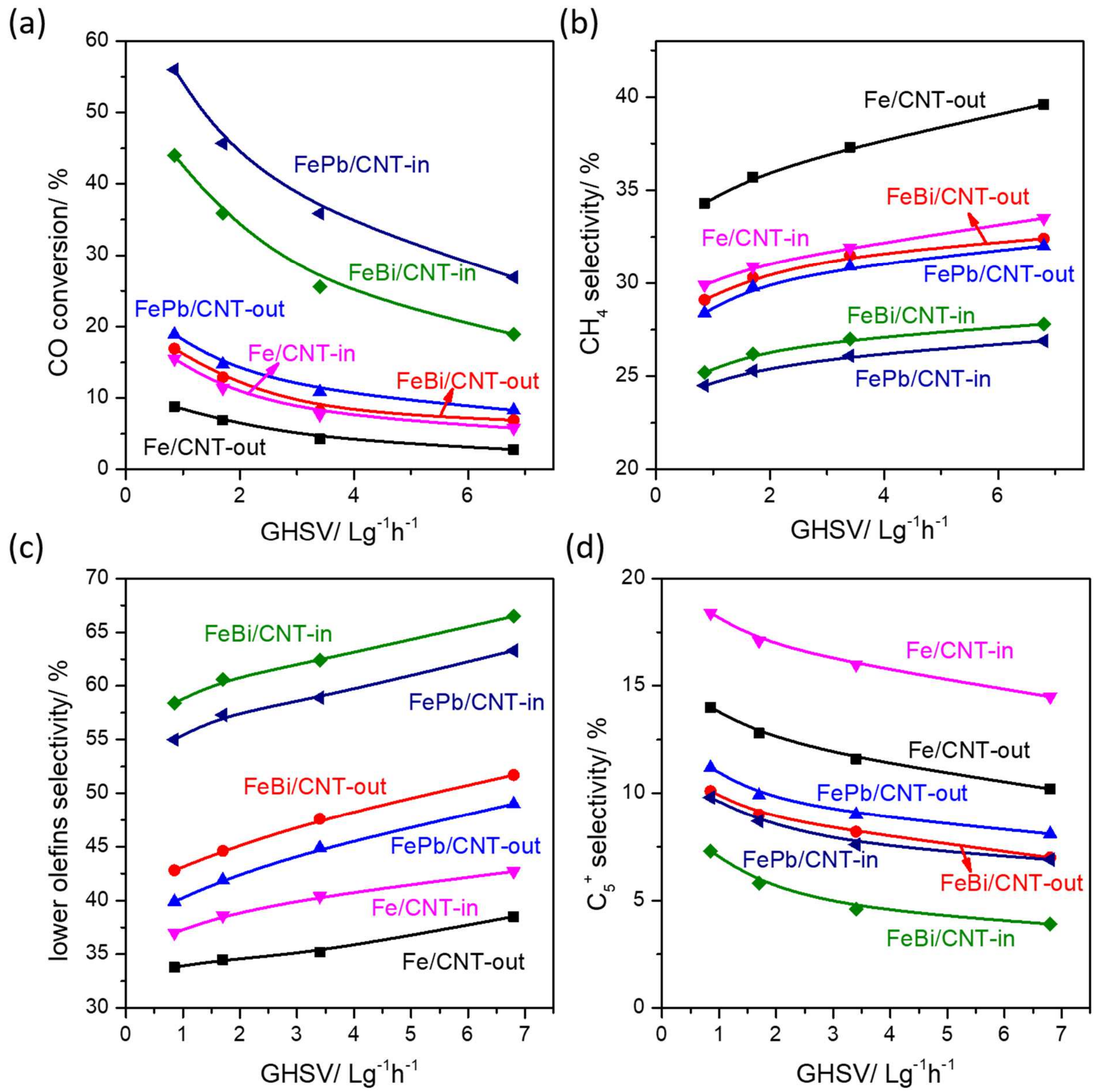


**Figure S8.** CO conversion and product selectivity versus GHSV over CNT non-confined and confined iron catalysts. Reaction conditions: $W$ = 0.1 g, $H_2/CO$ = 1, $P$ = 1 bar, $T$ = 350 °C, $GHSV$ = 0.85-6.8 L $h^{-1}g^{-1}$.

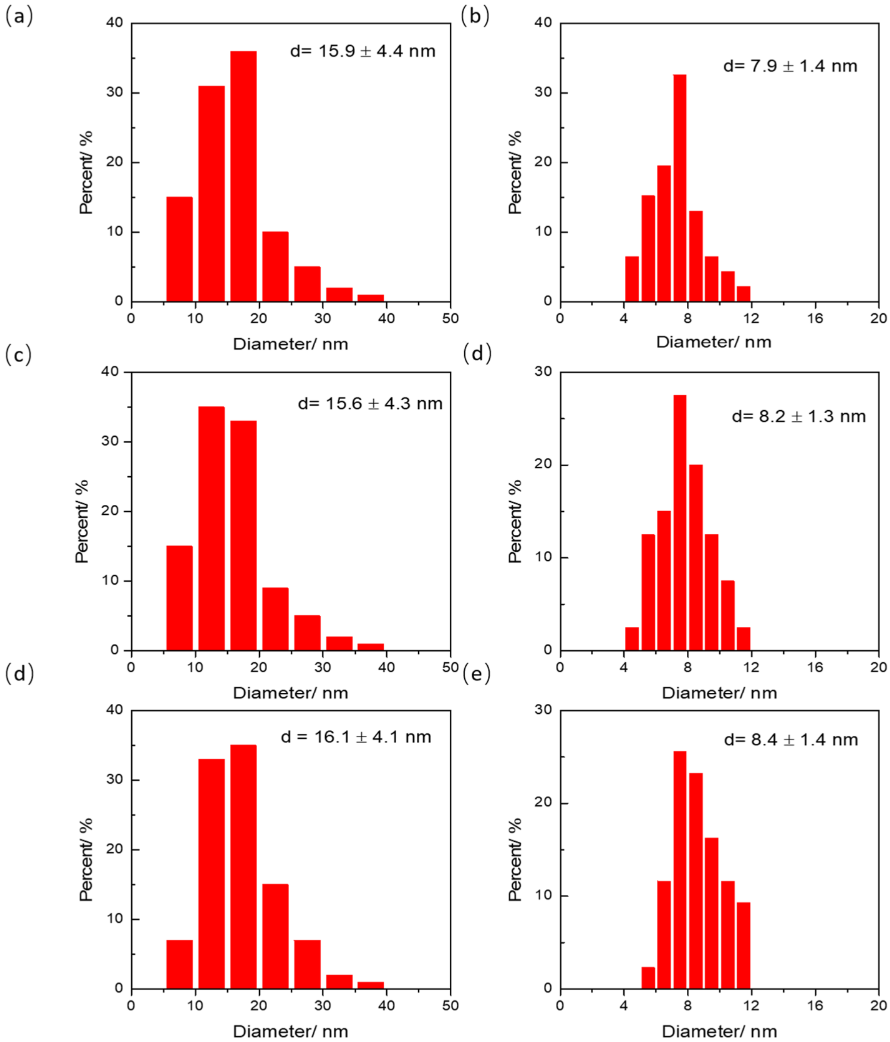


**Figure S9**. Particle size distribution for the used catalysts: (a) Fe/CNT-out-R, (b) Fe/CNT-in-R, (c) FeBi/CNT-out-R, (d) FeBi/CNT-in-R, (e) FePb/CNT-out-R, (f) FePb/CNT-in-R.

**Reference**

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